

Data Semantics – Assignments – 2

Ontology Modeling

Modify the **companies-geo.rdfs-owl-L2-2.owl** ontology with Protégé by creating two ontologies as specified below.

1. **companies-geo.rdfs-owl-L2-2-Your_Surname_clean.owl**

These specifications must be satisfied in ensemble in the final ontology; the order in which they are listed here below does not need to reflect the optimal order for a sequential implementation.

- Add two more branches to the ontology (the new classes are highlighted in bold):
 - **Person** > **Student**
 - Organization > **University**
- Make sure that each new class has at least two instances; you must appear as an instance of Student
- Model the following patterns by specifying the required object and datatype properties and by introducing two triples for each the pattern
 - Persons are friends to each other
 - Persons have a birthplace
 - Persons have names
 - Students attends universities
 - Universities have addresses
- Interpret the above patterns by specifying domain and range restrictions over object and datatype properties so as to implement intuitively sound inferences in the ontology
- Use a reasoner to make sure that there are at least two examples of desired inferences, based on your axiomatization

Provided that you satisfy the above specifications, you can make any additional change to the ontology needed to improve your ontology.

Remarks:

- At the exam, you may be asked questions about your ontology, e.g., to justify your choices

2. **companies-geo.rdfs-owl-L2-2-Your_Surname_wrong.owl**

Modify the **companies-geo.rdfs-owl-L2-2-Your_Surname_clean.owl** ontology in such a way that either an inconsistency is derived or an unintended (or, counterintuitive, or undesirable) inference is drawn.